

An Efficient Method to Estimate Admittance of Black-boxed Inverter-based Resources for Varying Operating Points

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Abstract—Traditional analytical approaches for stability assessment of inverter-based resources (IBRs), often requiring detailed knowledge of IBR internals, become impractical due to IBRs' proprietary nature. Admittance measurements, relying on electromagnetic transient simulation or laboratory settings, are not only time-intensive but also operationally inflexible, since various non-linear control loops make IBRs' admittance models operating-point dependent. Therefore, such admittance measurements must be performed repeatedly when operating point changes. To avoid time-consuming and cumbersome measurements, admittance estimation for arbitrary operating points is highly desirable. However, existing admittance estimation algorithms usually face challenges in versatility, data demands, and accuracy. Addressing this challenge, this letter presents a simple and efficient admittance estimation method for black-boxed IBRs, by utilizing a minimal set of seven operating points to solve a homogeneous linear equation system. Case studies demonstrate this proposed method ensures high accuracy across various types of IBRs. Estimation accuracy is satisfying even when non-negligible measurement errors exist.

Index Terms—Admittance estimation, black box, inverter-based resource, measurement error, operating point.

I. INTRODUCTION

INVERTER-BASED resources (IBRs) are increasingly replacing conventional synchronous generators to create a decarbonized modern grid [1], [2]. However, multi-timescale control interactions between IBRs and intricate grid conditions can trigger wideband instability issues, which can be identified by analyzing their admittance frequency responses (AFRs) [3], [4].

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The black-box nature of IBRs can impede theoretical derivation of their admittance models. Established admittance measurement techniques provide a means to address this challenge [5]. Moreover, recent methodological advancements have expanded these capabilities to ensure accurate admittance measurement even under adverse grid conditions such as voltage sags, imbalance, and harmonics [6]. However, obtained AFRs of IBRs are only valid at specific operating points defined by terminal voltage and output power/current, due to presence of non-linear power control, voltage control, and phase-locked loops (PLLs) [7]. It is thus necessary to repeatedly measure AFRs, even if there are slight variations in operating point. Given potentially infinite number of operating points in practice, development of an efficient admittance estimation method that can predict AFRs at any arbitrary operating points using only a few available admittance data is highly desirable.

The method proposed in [8], based on a nearly decoupled admittance model which decouples output current of a dc-voltage-controlled PLL-based grid-following inverter (GFLI) from its circuit and controller parameters, can estimate admittance using only ten admittance data points. However, terminal voltage still couples with internal parameters, leading to three sets of complicated measurement equations. Moreover, the method does not consider possible combinations of terminal voltages and output currents in admittance model, and is not generalized for PLL-less grid-forming inverters (GFMI) or other types of PLL-based GFLIs. Although improvements regarding the above three aspects were achieved in [9], the proposed method requires at least 28 admittance data points, which increases burden of acquiring data needed. An artificial neural network was employed in [10] to build the nonlinear relationship between admittance model and operating points. However, the training procedure needs a large number of admittance data points; non-negligible estimation error exists even if training data is accurate; and method is only verified on a dc-voltage-controlled GFLI similar to [8]. In addition, how to handle measurement errors has not been discussed in [8]–[10].

This letter presents a simple and efficient operating-point-dependent admittance estimation method (OAEM) for black-boxed IBRs that necessitates as few as seven admittance data points. The key to this method's efficiency is application

of a unified admittance model for IBRs, which simplifies the problem to solving a small-sized homogeneous linear equation system. Consequently, the method requires only as few as seven admittance data, significantly streamlining the estimation process without compromising accuracy. The remainder of this letter will formulate the investigated problem, outline development of the OAEM, and verify its effectiveness through case studies. Finally, conclusion is drawn.

II. PROBLEM FORMULATION

A. Black-box Feature of IBRs

Figure 1 presents a generalized black-boxed IBR system, where types and parameters of power filter and control system are unknown. The control system takes voltage $\bar{v}_{p,abc}$ at point of common coupling (PCC) and grid/converter current $\bar{i}_{g,abc}/\bar{i}_{t,abc}$ as inputs, while reference can be a power, voltage, or current command.

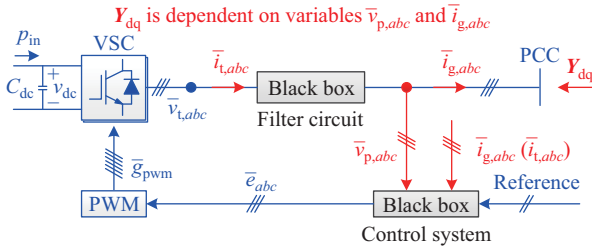


Fig. 1. An IBR with black-boxed circuit and control.

Admittance matrix $\mathbf{Y}_{dq}$ is dependent on operating point due to various non-linear control loops [7]. Moreover, varying output current of IBR can change PCC voltage via grid impedance [11], leading to a generalization of $\mathbf{Y}_{dq}$ as:

$$\mathbf{Y}_{dq} = \begin{bmatrix} y_{dd} & y_{dq} \\ y_{qd} & y_{qq} \end{bmatrix} = f(v_d, i_d, i_q, s) \quad (1)$$

where v_d represents d-axis PCC voltage; i_d and i_q represent d- and q-axis output currents, respectively, and s is Laplace operator. Note q-axis PCC voltage v_q is not considered in (1), since its value is kept at 0 V.

B. Unified Admittance Model for IBRs

The essential idea behind developed OAEM is each element of dq-domain admittance model of IBRs in (1) can be expressed in the form of [7]:

$$y_\varepsilon|_{s=j\omega_0} = \frac{\sum_{m=0}^M \sum_{i=0}^m v_d^i \left(\sum_{j=0}^{m-i} b_n^\varepsilon i_d^j i_q^{m-i-j} \right)}{\sum_{m=0}^M \sum_{i=0}^m v_d^i \left(\sum_{j=0}^{m-i} a_n^\varepsilon i_d^j i_q^{m-i-j} \right)} \quad (2)$$

where $\varepsilon \in \{dd, dq, qd, qq\}$; ω_0 is investigated frequency point; a_n^ε and b_n^ε represent coefficients of polynomial term $v_d^i i_d^j i_q^{m-i-j}$ in denominator and nominator, respectively, and pair (m, i, j) is indexed by

$$n = \frac{m(m+1)(m+2)}{6} - \frac{i(i-2m-3)}{2} + j \quad (3)$$

Additionally, M is the highest polynomial order for $v_d^i i_d^j i_q^{m-i-j}$ (i.e., $m \leq M$), leading to a total number of polynomial terms as $N = M(M^2 + 6M + 11)/6 + 1$.

Since multiplying the same non-zero constant on both the numerator and denominator of the right-hand side of (2) does not change $y_\varepsilon|_{s=j\omega_0}$, only $2N - 1$ of the $2N$ coefficients are independent of each other. Therefore, in this letter, a_0^ε will be fixed to a known value while $\bar{a}_\varepsilon = [a_1^\varepsilon, \dots, a_{N-1}^\varepsilon]^T \in \mathbb{C}^{(N-1) \times 1}$ and $\bar{b}_\varepsilon = [b_0^\varepsilon, \dots, b_{N-1}^\varepsilon]^T \in \mathbb{C}^{N \times 1}$ are to be calculated or optimized based on a number of measured data points. Then, admittance at another operating point can be estimated, which will be discussed in the next section.

III. DERIVATION OF THE PROPOSED OAEM

A. Manipulation of Available Admittance Data

Noting $a_0^\varepsilon \neq 0$, substituting K unique measured admittance data points into (2) leads to

$$\begin{aligned} & \sum_{m=0}^M \sum_{i=0}^m v_{d,k}^i \left(\sum_{j=0}^{m-i} y_{\varepsilon,k} i_{d,k}^j i_{q,k}^{m-i-j} a_n^\varepsilon \right) \\ & - \sum_{m=0}^M \sum_{i=0}^m v_{d,k}^i \left(\sum_{j=0}^{m-i} i_{d,k}^j i_{q,k}^{m-i-j} b_n^\varepsilon \right) = 0, \\ & \forall k \in \{1, \dots, K\} \end{aligned} \quad (4)$$

where $y_{\varepsilon,k}$ is measured admittance at k^{th} operating point $(v_{d,k}, i_{d,k}, i_{q,k})$.

Further, (4) can be compactly expressed as:

$$\mathbf{\Gamma} \bar{x}_\varepsilon = [\mathbf{\Gamma}_0, \mathbf{\Gamma}_1, \mathbf{\Gamma}_2] \times [\bar{a}_\varepsilon^T, \bar{b}_\varepsilon^T]^T = \bar{0}_K \quad (5)$$

where $\mathbf{\Gamma} = [\mathbf{\Gamma}_0, \mathbf{\Gamma}_1, \mathbf{\Gamma}_2]$, $\bar{x}_\varepsilon = [\bar{a}_\varepsilon^T, \bar{b}_\varepsilon^T]^T$, and $\bar{0}_K$ is K -dimensional zero vector.

For $\mathbf{\Gamma}_0 \in \mathbb{C}^{K \times 1}$, $\mathbf{\Gamma}_1 \in \mathbb{C}^{K \times (N-1)}$, and $\mathbf{\Gamma}_2 \in \mathbb{R}^{K \times N}$, they are defined as:

$$\mathbf{\Gamma}_0 = [y_{\varepsilon,1} \bar{t}_1^T, \dots, y_{\varepsilon,K} \bar{t}_K^T]^T_{(:,1:1)} = \mathbf{Y}_\varepsilon \mathbf{T}_1 \quad (6a)$$

$$\mathbf{\Gamma}_1 = [y_{\varepsilon,1} \bar{t}_1^T, \dots, y_{\varepsilon,K} \bar{t}_K^T]^T_{(:,2:N)} = \mathbf{Y}_\varepsilon \mathbf{T}_2 \quad (6b)$$

$$\mathbf{\Gamma}_2 = [-\bar{t}_1^T, \dots, -\bar{t}_K^T]^T = -\mathbf{T} = -[\mathbf{T}_1, \mathbf{T}_2] \quad (6c)$$

where $\bar{t}_k$ is defined as:

$$\bar{t}_k = [v_{d,k}^0 i_{d,k}^0 i_{q,k}^0, \dots, v_{d,k}^i i_{d,k}^j i_{q,k}^{m-i-j}, \dots, v_{d,k}^0 i_{d,k}^0 i_{q,k}^M]^T \quad (7)$$

$\mathbf{A}_{(:,k_1:k_2)}$ represents matrix consisting of columns k_1 to k_2 of a matrix $\mathbf{A}$; $\mathbf{Y}_\varepsilon = \text{diag}([y_{\varepsilon,1}, \dots, y_{\varepsilon,K}]^T)$ and $\mathbf{T} = [\bar{t}_1^T, \dots, \bar{t}_K^T]^T$ are matrices in $\mathbb{C}^{K \times K}$ and $\mathbb{R}^{K \times N}$, respectively; $\mathbf{T}_1$ is first column of $\mathbf{T}$, while $\mathbf{T}_2$ is a matrix consisting of columns 2 to N of $\mathbf{T}$.

Combining (5) and (6) leads to

$$\mathbf{Y}_\varepsilon \mathbf{T}_1 \bar{a}_\varepsilon + \mathbf{W} \begin{bmatrix} \bar{a}_\varepsilon \\ \bar{b}_\varepsilon \end{bmatrix} = 0 \quad (8)$$

where $\mathbf{W} = [\mathbf{Y}_\varepsilon \mathbf{T}_2, -\mathbf{T}_1, -\mathbf{T}_2]$.

Then, optimal values for $\bar{a}_\varepsilon^*$ and $\bar{b}_\varepsilon^*$ can be calculated as:

$$\begin{bmatrix} \bar{a}_\varepsilon^* \\ \bar{b}_\varepsilon^* \end{bmatrix} = -(\mathbf{W}^T \mathbf{W})^{-1} \mathbf{W}^T \mathbf{Y}_\varepsilon \mathbf{T}_1 \quad (9)$$

where $\bar{a}_\varepsilon^*$ and $\bar{b}_\varepsilon^*$ are calculated or optimized values for $\bar{a}_\varepsilon$ and $\bar{b}_\varepsilon$, respectively.

B. Estimating Admittance for Another Operating Point

Assuming an operating point $(v_d^{\text{pr}}, i_d^{\text{pr}}, i_q^{\text{pr}})$, $\bar{t}_{\text{pr}}$ can be built first as:

$$\bar{t}_{\text{pr}} = [(v_d^{\text{pr}})^0 (i_d^{\text{pr}})^0 (i_q^{\text{pr}})^0, \dots, (v_d^{\text{pr}})^i (i_d^{\text{pr}})^j (i_q^{\text{pr}})^{m-i-j}, \dots, (v_d^{\text{pr}})^0 (i_d^{\text{pr}})^0 (i_q^{\text{pr}})^M]^T \quad (10)$$

Then, substituting (10) into (2) leads to

$$y_{\varepsilon}^{\text{pr}}|_{s=j\omega_0} \bar{t}_{\text{pr}}^T [\bar{a}_{\varepsilon}^{\varepsilon}; \bar{a}_{\varepsilon}^*] - \bar{t}_{\text{pr}}^T \bar{b}_{\varepsilon}^* = 0 \quad (11)$$

where $y_{\varepsilon}^{\text{pr}}|_{s=j\omega_0}$ is admittance to be estimated, leading to

$$y_{\varepsilon}^{\text{pr}}|_{s=j\omega_0} = \frac{\bar{t}_{\text{pr}}^T \bar{b}_{\varepsilon}^*}{\bar{t}_{\text{pr}}^T [\bar{a}_{\varepsilon}^{\varepsilon}; \bar{a}_{\varepsilon}^*]} \quad (12)$$

Requirement to solve only a small-sized homogeneous linear equation system highlights method's computational efficiency. Moreover, the model-based approach contributes to its capability for accurate admittance estimations. Several remarks are given below.

1) $\bar{a}_{\varepsilon}^*$ and $\bar{b}_{\varepsilon}^*$ can be calculated with arbitrary number of data points. However, accuracy is expected to be higher if more data points are available.

2) Since $\text{rank}(W) \leq \min\{K, 2N - 1\}$, significant errors may be introduced due to incomplete information provided (insufficient data points) when $K < 2N - 1$. By contrast, the least square method (LSM) is implicitly used in seeking optimal values for $\bar{a}_{\varepsilon}$ and $\bar{b}_{\varepsilon}$ when $K > 2N - 1$. Therefore, ideally, number of available data points should be no less than $2N - 1$ in order to ensure $\bar{a}_{\varepsilon}^*$ and $\bar{b}_{\varepsilon}^*$ are with acceptable accuracy.

3) The proposed approach can also handle the case when measurement errors exist in available data points, where LSM will be used by (9) if more than $2N - 1$ data points are available.

IV. CASE STUDY

A. Available Admittance Data without Measurement Error

With $M = 1$, $K = 2N - 1 = 7$ qq-axis 1-Hz admittance

data of current- (#1 GFLI), voltage- (#2 GFLI), and power-controlled (#3 GFLI) GFLIs, are calculated by the method presented in [7] and are listed at top of Table I. The parameters v_d , i_d , and i_q are limited within $[0.9, 1.1]$, $[0, 1.0]$, and $[-1.0, 1.0]$ p.u., respectively, and power capacity is constrained by 1.0 p.u. It is noteworthy v_d is contingent upon i_d and i_q subject to prevailing grid conditions. Diverse combinations of v_d , i_d , and i_q delineated in Table I encapsulate a broad spectrum of operational scenarios, effectively emulating the dynamic nature of grid impedance. This methodical selection of these scenarios ensures admittance estimation method is thoroughly evaluated against a generalized set of grid conditions rather than a limited or static impedance profile. qq-axis 1-Hz admittance data at four operating points @1-@4 are estimated by the proposed OAEM and listed at the middle of Table I, which are in line with theoretical admittance data listed at bottom of Table I. Fig. 2(a), (b), and (c) further show 2-, 5-, 10-, 40-, and 100-Hz qq-axis admittance data at the four operating points are successfully estimated for all types of GFLIs.

With $M = 2$, $K = 2N - 1 = 19$ admittance data points are available for VSG (#4 GFMI). Of the 19 data points, seven of them are listed at top of Table I for brevity, while estimated admittance values are listed at middle of the table. It is evident estimated values are extremely close to their true values listed at bottom of the table. Moreover, Fig. 2(d) also demonstrates admittance data at other frequencies, including 2-, 5-, 10-, 40-, and 100-Hz, are also accurately estimated.

Note the proposed OAEM can seamlessly be extended to estimate admittance of remaining three admittance components. Given operating point variations predominantly influence low-frequency admittance characteristics due to limited bandwidths of nonlinear control loops, as evidenced in Fig. 2, this letter focuses on admittance estimation within low-frequency domain. However, the proposed OAEM is designed to estimate admittance at each individual frequency point, which makes it robust and extendable to a broader frequency range. This attribute ensures OAEM can be effectively applied in wider-band stability analysis, which is vital for practical handling of IBRs in various operational scenarios. Notably, the proposed OAEM

TABLE I
IMPLEMENTATION RESULTS ON THE QQ-AXIS 1-HZ ADMITTANCE DATA

(v_d, i_d, i_q) (p.u.)	#1 GFLI	#2 GFLI	#3 GFLI	#4 GFMI
Seven available qq-axis 1-Hz admittance data				
(0.92, 0.1, 0.6)	1.08∠183.80	1.25∠174.92	1.26∠177.91	4.13∠−107.67
(0.93, 0.2, 0.5)	2.07∠181.86	2.43∠173.45	2.45∠176.32	5.10∠−112.30
(0.95, 0.3, −0.4)	3.01∠181.20	3.52∠173.12	3.57∠175.96	7.18∠75.33
(0.97, 0.4, −0.3)	3.90∠180.87	4.57∠173.04	4.62∠175.86	9.46∠67.22
(1.03, 0.5, 0.2)	4.58∠180.70	5.31∠173.41	5.37∠176.23	9.07∠−101.35
(1.05, 0.7, −0.2)	6.26∠180.44	7.25∠173.39	7.34∠176.20	12.88∠58.31
(1.08, 0.9, 0.1)	7.81∠180.31	9.00∠173.51	9.11∠176.32	12.19∠−89.65
Four estimated qq-axis 1-Hz admittance data				
@1(1.00, 0.3, 0.3)	2.86∠181.28	3.32∠173.59	3.35∠176.44	6.98∠−105.34
@2(1.10, 0.3, 0.3)	2.61∠181.43	2.96∠174.40	2.99∠177.25	6.51∠−106.52
@3(1.00, 0.6, 0.3)	5.64∠180.52	6.59∠173.04	6.67∠175.85	8.27∠−117.21
@4(1.00, 0.3, 0.6)	2.86∠181.28	3.32∠173.59	3.35∠176.44	4.81∠−125.10
Four theoretical qq-axis 1-Hz admittance data at the estimated operating points				
@1(1.00, 0.3, 0.3)	2.86∠181.28	3.32∠173.59	3.35∠176.44	6.98∠−105.34
@2(1.10, 0.3, 0.3)	2.61∠181.43	2.96∠174.40	2.99∠177.25	6.51∠−106.52
@3(1.00, 0.6, 0.3)	5.64∠180.52	6.59∠173.04	6.67∠175.85	8.27∠−117.21
@4(1.00, 0.3, 0.6)	2.86∠181.28	3.32∠173.59	3.35∠176.44	4.81∠−125.10

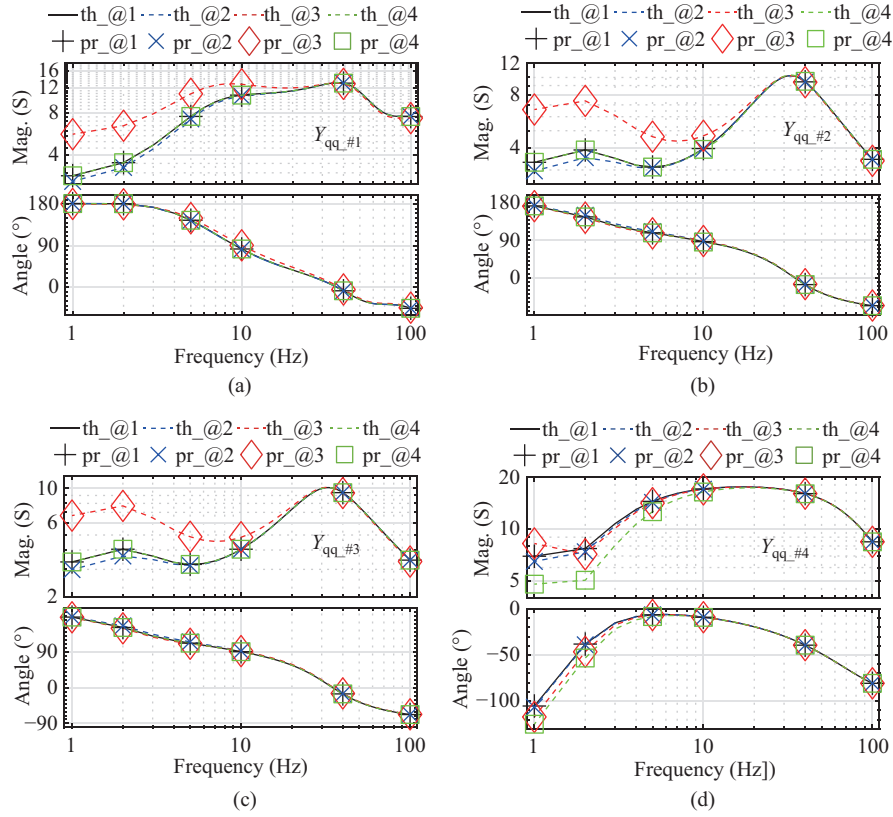


Fig. 2. Implementation of the proposed OAEM on qq-axis admittance of (a) #1 current-controlled GFLI, (b) #2 voltage-controlled GFLI, (c) #3 power-controlled GFLI, and (d) #4 VSG-controlled GFMI using ideal theoretical admittance data points.

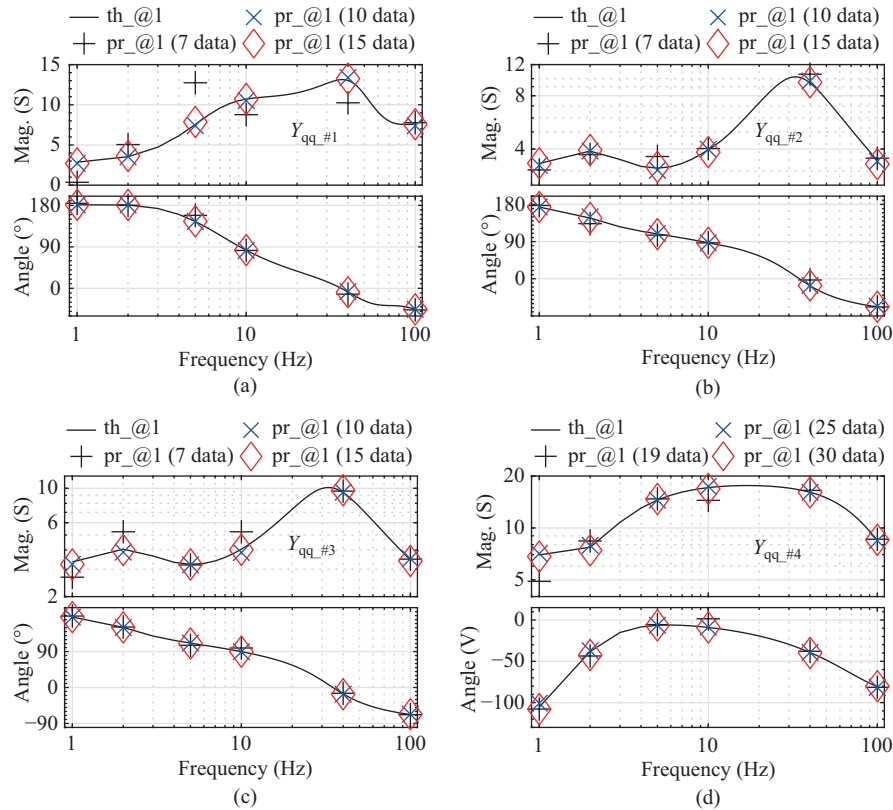


Fig. 3. Implementation of the proposed OAEM on qq-axis admittance of (a) #1 current-controlled GFLI, (b) #2 voltage-controlled GFLI, (c) #3 power-controlled GFLI, and (d) #4 VSG-controlled GFMI using admittance data points polluted by 5% white noises.

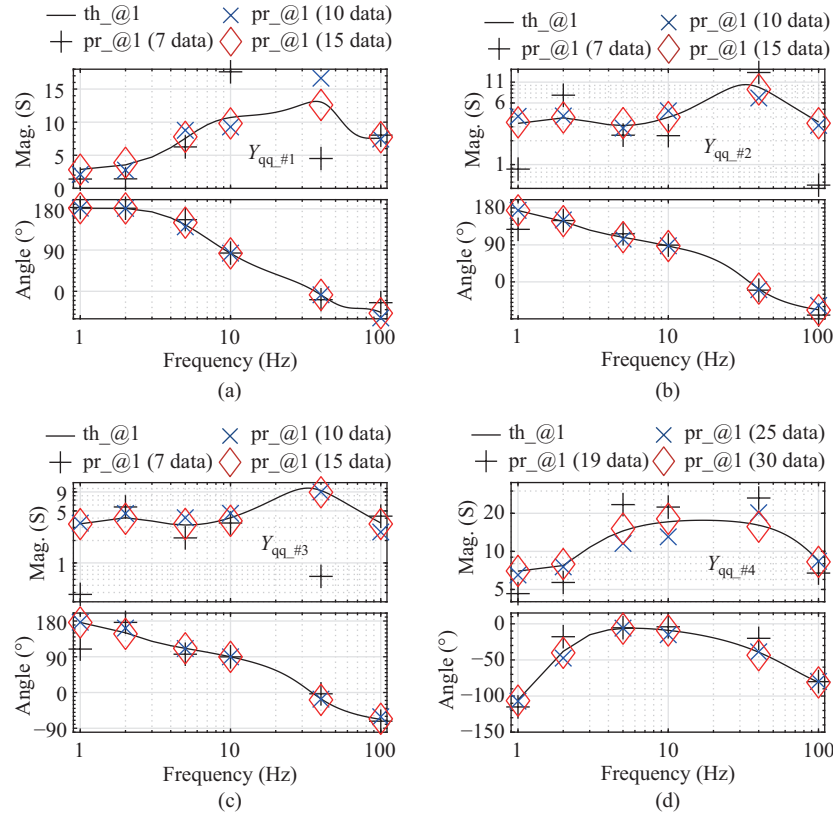


Fig. 4. Implementation of the proposed OAEM on qq-axis admittance of (a) #1 current-controlled GFLI, (b) #2 voltage-controlled GFLI, (c) #3 power-controlled GFLI, and (d) #4 VSG-controlled GFMI using admittance data points polluted by 10% white noises.

estimates admittance on a frequency-by-frequency basis. This granularity ensures any arbitrary frequency, including resonance frequency points (e.g., 2 Hz as showcased in Fig. 2(c)), can be estimated with high precision. It's worth noting while the GFLIs necessitate at least seven admittance data, the GFMI prescribes a minimum of 19. Therefore, when implementing a practical black-box IBR, employing 19 admittance data is recommended for reliable admittance estimation.

B. Available Admittance Data with Measurement Errors

To assess reliability of the proposed OAEM when measurement errors exist, white noise at levels of 5% and 10% was incorporated into conductance and susceptance values of available data points. Impact of noise on the OAEM's estimation accuracy is presented in Fig. 3 for 5% noise scenario and in Fig. 4 for 10% case. These figures demonstrate estimation precision is influenced by quantity of data points, with limited points (7 for GFLIs and 19 for GFMI) leading to reduced accuracy. However, as number of data points increases, a significant improvement in estimation precision is observed.

V. CONCLUSION

This letter develops an admittance estimation method for black-boxed IBRs. This method was shown to be simple, efficient, and capable of estimating AFRs at arbitrary operating points using as few as seven admittance data, thereby avoiding repeated and time-consuming admittance measurements. Additionally, robustness of the proposed OAEM against non-negligible measurement errors is also demonstrated.

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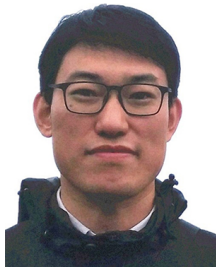
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